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The Hyperon–Nucleon Interaction in Effective Field Theory

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Contents

Contents	i
Introduction	1
1 Short review of Effective Field Theory approaches	4
1.1 The EFT approach in nuclear physics	4
1.2 Outlook on the literature that guided our work	6
2 Re-fitting the ΛN contact interaction	8
2.1 Overview of our previous work	8
2.2 New fitting procedure developed in this project	10
Numerical implementation of the fitting procedure 11 • Analysis and selection of the fit results 12	
2.3 Results of the fitting procedure.	13
Fit quality and regulator dependence 14 • High-energy predictions 15 • Overall considerations 15	
3 Work-in-progress: applications and extensions of the ΛN interaction	17
3.1 Application to light hypernuclei	17
3.2 Extension to a local chiral interaction	18
4 Conclusions and future developments	19
References	20

I Introduction

In recent years, there has been a notable interest in investigating hyper-nuclear systems, *i.e.* nuclear systems where one or more nucleons (N), which is the generic way of referring to neutrons and protons, are replaced by a hyperon (Y). Hyperons are baryons containing one or more strange quarks, they are classified as Fermions, and their lifetimes are of the order of 10^{-10} s. Hyperons with one strange quark include the Λ and the Σ triplet, *i.e.*, Σ^+ , Σ^0 , Σ^- , while those with two strange quarks include the Ξ^0 and Ξ^- . Hyperons can form hypernuclei, which are nuclei where one or more nucleons are substituted by these baryons. Recent experimental efforts have led to the identification of over 40 single- Λ hypernuclei and a few double- Λ hypernuclei. Nevertheless, the existence of single- Σ hypernuclei remains unconfirmed, indicating that the interaction involving Σ hyperons is likely repulsive [1]. In this study, we paid particular attention to the Λ hyperon, which has a mass of (1115.683 ± 0.006) MeV [2], and whose total isospin and charge are both equal to 0. Although the Λ lifetime is $(2.617 \pm 0.010) \times 10^{-10}$ s [2], in some particular conditions, as inside a neutron star (NS), it can become a stable particle. The existence of hyperons within a NS represents a topic of considerable debate [1] [3]. This is mainly related to the so-called “hyperon puzzle”, which we will briefly discuss later: we need first to discuss the formation of a NS.

The formation of a NS typically occurs due to the gravitational collapse of a massive progenitor star, which has ceased nuclear fusion reactions. An alternative formation mechanism for a NS is through a binary NS merger. This process was observed for the first time with the gravitational wave signal of event GW170817 [4].

For progenitor stars with masses between approximately 8 and 30 solar masses ($M_\odot$)¹, the evolutionary process involves nuclear fusion up to iron (^{56}Fe), the most stable element. This fusion process creates an onion-like structure with heavier elements in the core and lighter elements in the outer layers. As the core density increases, the process of inverse beta decay, $e^- + p \rightarrow n + \nu_e$, becomes more efficient, converting electrons and protons into neutrons and neutrinos. While the iron core forms, nuclear reactions temporarily sustain the star equilibrium. However, once the core mass exceeds a critical limit, gravity overcomes the internal pressure, causing the core to collapse to densities of the order of 10^{14} g/cm³. This collapse generates a shock wave that expels most of the star outer material, resulting in a Supernova explosion [5], and leaves only a core mainly made of neutrons. If the remaining core has a mass between 2 and 3 $M_\odot$, it becomes a NS; otherwise, if the mass is greater, it forms a black hole. This mass limit that determines whether a Supernova remnant evolves into a NS or a black hole is known as the Tolman-Oppenheimer-Volkoff (TOV) mass limit. The presence of this limit is due to the Pauli exclusion principle, which applies to neutrons as Fermions and leads to degeneracy pressure. This degeneracy pressure is a crucial factor in maintaining the stability of the NS. If the mass of the Supernova remnant exceeds the TOV mass

¹Within this mass range, the star fate diverges into two possible outcomes: if the progenitor mass is between 8 and 20 $M_\odot$, the star will likely collapse into a NS; however, if the mass exceeds about 20 $M_\odot$, a black hole is probably formed [5]. Whether a black hole actually forms depends not only on the progenitor mass but also on other factors, such as instabilities that may arise during the dynamic collapse process.

limit, the degeneracy pressure is insufficient to prevent further gravitational collapse. It is important to note that the values of quantities such as the TOV mass limit and the central density of the NS depend on the model used for the Equation of State (EoS), which relates pressure to energy density for a given temperature and composition inside the NS. The EoS depends on the underlying interactions among nucleons and, if present, hyperons, which contribute to the energy density and pressure of the matter [3]. Additionally, the EoS also influences the outcome of binary NS mergers, which in some cases can lead to black hole formation.

A NS typically has a mass between 1.4 and 2.01 $M_{\odot}$ ² and a radius of approximately 10 to 12 kilometers [6]. The central densities of NSs can be of the order of $(2-3) \times 10^{14} \text{ g/cm}^3$ [5]. These high-density conditions lead to an increase in the Fermi energy of the nucleons due to the Pauli exclusion principle, making the conversion of nucleons into hyperons energetically favorable. This conversion results in a decrease in energy and, consequently, a reduction in pressure. This reduction in pressure is commonly referred to as the softening of the EoS, meaning that the pressure becomes less sensitive to changes in energy density within the NS. The softening of the EoS leads to an underestimation of the TOV mass, which contradicts experimental evidence. Thus, while the presence of hyperons in NSs appears unavoidable, their inclusion results in a TOV mass that is not consistent with measured NS masses. This discrepancy is known as the aforementioned “hyperon puzzle”. In order to solve this puzzle within nuclear physics, it is crucial to investigate the underlying causes of this phenomenon. This investigation involves a thorough understanding of the interactions between a hyperon and a nucleon (YN), the interactions between two hyperons (YY), and the three-body interactions such as YNN , YYN , and YYY [7].

Beyond the “hyperon puzzle”, a deeper insight into the YN interactions is important for further testing our knowledge of strong interactions, including the strange quark sector. For example, including hypernuclei in the table of nuclides³ would be desirable, as it would allow a more comprehensive understanding of the strangeness properties. Another valuable reason for studying this interaction from a theoretical perspective is that it would allow theoretical predictions to be compared with experimental results. The experimental campaign conducted by the ALICE Collaboration at the Large Hadron Collider (LHC) is particularly relevant in this context, as it provides a unique opportunity to study heavy-ion collisions that produce hyperons [8]. In particular, precise measurements of the correlation function⁴ for the Λp system have been obtained by the ALICE Collaboration [9]. These new data allow for a more precise analysis of the YN interaction, with respect to the scarce and outdated datasets involving Λp elastic cross section data.

Considering the importance of deepening our understanding of the YN interaction, this

²One of the most accurate measurements of NS masses was obtained with PSR J0348+0432, which has turned out to be $(2.01 \pm 0.04) M_{\odot}$. An interesting but slightly larger result was obtained from the analysis of the GW170817 event, where the inferred mass of one of the NS involved in the binary NS merger is about $2.3 M_{\odot}$.

³The table of nuclides is a two-dimensional representation of all known atomic nuclei, arranged by the number of protons (atomic number) and neutrons (neutron number). It provides a comprehensive overview of the nuclear properties and characteristics of these nuclides. Adding hypernuclei to this chart would involve adding a third axis to the existing chart, to represent strangeness.

⁴The correlation function quantifies how the emission of one particle affects the emission of another, providing insights into the underlying dynamics of the collision events.

project aims to study a local potential model for the ΛN interaction. By “local” model, we mean that the interaction is expressed in coordinate space and depends only on the relative distance between the particles, besides their spins and isospins. The potential has been constructed within an Effective Field Theory (EFT), based on Quantum Chromodynamics (QCD) symmetries, which includes only contact terms, *i.e.* terms where the nucleons interact in the same position in space-time, ultimately depending only on some free parameters that need to be fixed through fits to experimental scattering data.

The derivation of this potential and a first fitting procedure were developed in a previous work by our group, in Ref. [10]. Here, the derivation of the potential is left unchanged and only briefly summarized. The main goal is instead to develop a substantially more sophisticated and comprehensive fitting procedure, yielding a markedly more robust and reliable determination of the free parameters, as well as to study some applications of this model.

The one presented here is the first study where a local ΛN interaction is derived within an EFT approach⁵. The chosen framework, *i.e.* the EFT approach with only contact terms, offers several advantages: (i) EFT allows to perform a perturbative expansion and therefore to obtain a robust estimate of the theoretical errors deriving from the exclusion of higher order contributions; (ii) the presence of only contact terms leads to a simplified operatorial structure; (iii) working in coordinate space allows to derive a model that is easy to handle with numerical methods, especially those that involve massive hypernuclei.

This work is organized as follows: Chapter 1 briefly introduces EFT and reviews the main results in literature that guided our work. Chapter 2 summarizes the main features of the ΛN contact model we derived in our previous work, of Ref. [10], then it presents the new fitting procedure, its numerical implementation, along with the results. Chapter 3 introduces some “work-in-progress” projects: in particular, it discusses the application to the hypertriton and a different EFT approach, that allows to derive a more complete interaction. The final chapter summarizes the results and presents future developments of the “work-in-progress” projects.

⁵Other local potential models are phenomenological, which means that are not rooted to the underlying theory of QCD. To our knowledge, the only other local, contact-only EFT treatment of YN interactions is that of Ref. [11]. That model includes fewer interaction operators and is therefore less complete than the one considered here.

1 Short review of Effective Field Theory approaches

The aim of this chapter is to introduce the Effective Field Theory (EFT) formalism and explain how it connects low-energy nuclear physics with Quantum Chromodynamics (QCD). The discussion is intended as a brief and general introduction to the main concepts and terminology of EFT. Section 1.1 presents an overview of the EFT approach and its applications to nuclear physics, emphasizing the advantages of this framework. It also introduces two specific formulations: Chiral Effective Field Theory (χ EFT) and Pionless Effective Field Theory (π EFT). Both can be employed to describe interactions between nucleons and can be extended to more general baryon interactions. The present study focuses exactly on the extension to the hyperon–nucleon sector, which is discussed in Section 1.2, together with a brief review of the literature results on which this work is based.

1.1 The EFT approach in nuclear physics

QCD is the theory of strong interactions, that is, the interactions between quarks, which are the fundamental constituents of hadrons. These interactions are mediated by exchanges of gluons. Consequently, quarks and gluons represent the fundamental degrees of freedom of QCD. The strength of the interaction between quarks and gluons is characterized by the QCD coupling, a dimensionless parameter that depends on the momentum scale at which the interaction is probed. The QCD coupling decreases at high momentum scales, well above 1 GeV (corresponding to short distances), and becomes large at low momentum scales, associated with distances of order 1 fm or larger, as encountered in nuclear physics [12] [13]. This large coupling prevents a perturbative treatment of QCD in the low-energy regime: this means that the relevant physical quantities cannot be expanded in terms of a small coupling constant. As a result, the derivation of nuclear forces from QCD represents a significant challenge [13].

An EFT approach, first presented by Weinberg in Ref. [14], is a powerful tool for addressing this issue. This formalism is essentially a “low-energy” reduction of QCD, where the effective degrees of freedom are first of all nucleons, rather than quarks and gluons. The other degrees of freedom involved, besides nucleons, depend on the chosen approach. In particular, in χ EFT, valid in an energy regime up to about 100 MeV, the other degrees of freedom are pions (π), the mediators of nuclear interaction [13]. In an energy regime even lower, up to about 10 MeV, even pions can be considered as high-energy degrees of freedom, thus they can be integrated out, leaving only nucleons as the relevant ones. The EFT that describes this energy regime is known as π EFT [15].

In order to implement an EFT from QCD, it is necessary to construct Lagrangians that describe the interactions between the degrees of freedom of the theory. For example, in the context of χ EFT, the Lagrangian can be arranged as follows [13]:

$$\mathcal{L} = \mathcal{L}_{\pi\pi} + \mathcal{L}_{\pi N} + \mathcal{L}_{NN} + \dots, \quad (1.1)$$

where $\mathcal{L}_{\pi\pi}$ represents the interactions among two pions, $\mathcal{L}_{\pi N}$ describes the dynamics between a pion and a nucleon and $\mathcal{L}_{NN}$ is referred to two-nucleon interactions. The ellipsis stands for terms involving interactions between two or more nucleons, with or without pions. These Lagrangians must include all terms allowed by the symmetries of QCD, in particular chiral symmetry and its pattern of spontaneous and explicit breaking.

The effective Lagrangians obtained in this way have infinite terms, from which an infinite number of Feynman graphs can be calculated. To systematically organize the terms of the Lagrangian and the corresponding Feynman diagrams, a method called “power counting” is employed. The expansion is arranged in powers of the dimensionless parameter Q/Λ , where Q represents the low-energy scales of the process, such as the typical momentum or the pion mass in χ EFT, and Λ denotes the hard, or breakdown, scale above which the EFT is no longer valid. In the context of χ EFT, this scale is conventionally denoted by Λ_χ and has a value of approximately 1 GeV.¹ In the context of π EFT, instead, Q denotes low-momentum scales such as the typical relative momentum, inverse large scattering lengths, or binding momenta of shallow bound states, while $\Lambda \sim m_{\pi_0}$ [15]. Since Q/Λ is small within the domain of validity of the theory, terms with lower powers of this ratio are expected to provide the dominant contributions, while terms with progressively higher powers give smaller corrections. Power counting therefore provides both a prescription for truncating the otherwise infinite Lagrangian and a natural hierarchy for the resulting interaction. The lowest non-vanishing contribution is called leading order (LO), the first correction next-to-leading order (NLO), followed by next-to-next-to-leading order (N²LO), and so on.

Besides the merit of rooting the derivation of the nuclear interaction on QCD, the EFT approach also has the advantage of allowing a robust and systematic estimation of theoretical errors arising from truncating the expansion at a given order, corresponding to the quantity $(Q/\Lambda_\chi)^{\nu+1}$ when the theory is truncated at the ν -th order. This estimate is meaningful only for $Q \ll \Lambda$, in fact, as Q approaches the breakdown scale, the hierarchy between successive orders is lost and the EFT can no longer be expected to provide an accurate description.

Although the EFT is intended to describe only low-energy physics, loop integrals and the iteration of nuclear potentials can include arbitrarily large momenta. These contributions lie outside the domain of the expansion and can lead to high-momentum divergences. They are therefore suppressed through regularization, by multiplying the interaction by a smooth regulator function. A possible choice is a Gaussian regulator,

$$f_{\Lambda_{\text{reg}}}(q) = \exp\left[-\left(\frac{q}{\Lambda_{\text{reg}}}\right)^2\right] = \exp\left(-\frac{R_0^2 q^2}{4}\right), \quad \Lambda_{\text{reg}} = \frac{2}{R_0}, \quad (1.2)$$

here q is the transferred momentum and we use $\hbar \cdot c = 1$. The regulator cutoff Λ_{reg} , or equivalently the coordinate-space cutoff parameter R_0 , is a parameter and should not be confused with the physical breakdown scale Λ . The former controls which high-momentum contributions are suppressed in a calculation, whereas the latter determines the range of validity of the EFT expansion. This Gaussian form is the one adopted in the

¹The construction of an EFT requires a clear separation between the soft and hard scales. In χ EFT, this separation reflects the small pion mass and external momenta compared with the masses of heavier hadrons [13].

present work, following Ref. [16], which guided the derivation of our interaction. This choice is further motivated by the fact that a Gaussian regulator is local in coordinate space, *i.e.*, it depends only on the relative distance between the two interacting particles, which is a property we required for our model.

The regulator prevents the theory from resolving short-distance physics beyond its range of applicability, but it also introduces a dependence on the chosen function and cutoff. The effects of this unresolved high-energy dynamics are accounted for by contact terms, whose strengths are encoded in a set of free parameters known as Low Energy Constants (LECs), introduced through a procedure known as renormalization. At each order, power counting determines which contact operators, and correspondingly which LECs, must be included. The LECs are then fixed by fitting low-energy experimental data, typically scattering observables and binding energies, and in doing so they absorb the regulator dependence, varying with the chosen cutoff. Physical observables, by contrast, should remain approximately cutoff independent, up to corrections from orders omitted in the expansion [13].

1.2 Outlook on the literature that guided our work

The results summarized in this section are the starting point for the interaction developed in the present work. The main idea is to follow the π EFT approach used in Ref. [16] to derive a contact and local nucleon–nucleon interaction, but to apply it to the hyperon–nucleon sector. For this purpose, we combine elements from different references.

As already mentioned, in Ref. [16] the π EFT formalism is applied to the nucleon–nucleon sector. In π EFT, pions are not explicit degrees of freedom. Only the nucleons remain, and the effects of pion exchange and of shorter-range physics are included in contact terms. The interaction of Ref. [16] therefore contains contact terms only. These terms are multiplied by Gaussian regulators that depend on the transferred momentum, as discussed in the previous section. This choice gives a local interaction in coordinate space. In Ref. [16], the authors derive the interaction up to N³LO, while in our work we limit ourselves to NLO. We adopt Gaussian regulators for our ΛN interaction. To be consistent with this derivation, we also included the same values of the cutoff parameter R_0 , which are in the range $R_0 \in [0.7, 2.5]$ fm.

The extension from the nucleon–nucleon to the hyperon–nucleon sector requires the inclusion of additional degrees of freedom, because in this case also the strange quark (besides up and down quarks) is taken into account. The relevant baryonic degrees of freedom are then the nucleons and the Λ , Σ , and Ξ hyperons. The exchanged pseudoscalar mesons are the π , K , and η .

To our knowledge, the only references that have developed a hyperon–nucleon interactions are Refs. [17] [18] [19] [20] and [11]. The latter is the only one involving a local interaction (in coordinate space) with contact terms only (π EFT), but it is limited to LO. All the other references use a χ EFT approach, thus including both contact terms and explicit one- and two-meson-exchange contributions. In particular, Ref. [17] derives the interaction up to LO, Refs. [18] [19] up to NLO, and Ref. [20] up to N²LO. In all these references, the interaction is formulated in momentum space and uses a non-local regularization scheme (*i.e.* the regulator depends separately on the initial and final

momenta of the interacting particles). The only exception is the potential of Ref. [20], where a semilocal regularization is used. In this scheme, a local regulator is applied to the meson-exchange contributions, while the contact terms are regulated non-locally. The complete interaction is therefore also non-local. Its LECs are fitted mainly to the available low-energy YN scattering data (which are the same experimental data used for the fitting procedure in our work), with additional information from hypernuclei, such as the hypertriton binding energy.

Our construction takes the contact operator structure up to NLO from the chiral YN interaction of Ref. [20], but removes the explicit meson-exchange terms. In this sense, we apply the main idea of the π EFT of Ref. [16] to the YN sector: at sufficiently low energies, the effects of meson exchange are not described explicitly but are absorbed into the LECs of contact operators. We then regularize these operators locally in coordinate space, following Ref. [16], and determine their LECs from low-energy Λp scattering data. The whole derivation of our model and a first attempt to fit the LECs were performed in Ref. [10]. In this project, and specifically in Chapter 2, we present a new and more sophisticated fitting procedure and then present a more complete analysis of the results.

Re-fitting the ΛN contact interaction

This chapter presents the main project of this thesis: a new fitting procedure for the local ΛN contact interaction developed in Ref. [10]. The derivation of the potential and the calculation of the elastic Λp cross section are left unchanged and are briefly reviewed first. We then describe the new numerical strategy used to determine the LECs and discuss the resulting fit quality, regulator dependence, and multiplicity of solutions.

2.1 Overview of our previous work

The ΛN contact potential model

In Ref. [10], we applied the same approach as the one presented in Ref. [16] to the χ EFT potential model for the hyperon-nucleon interaction derived in Ref. [20]. The final expression for our model in coordinate space, up to NLO, is

$$V_{\Lambda N}^{\text{LO}} = [C_S + C_T(\boldsymbol{\sigma}_\Lambda \cdot \boldsymbol{\sigma}_N)]F(r), \quad (2.1)$$

$$\begin{aligned} V_{\Lambda N}^{\text{NLO}} = & [C_1 + C_2(\boldsymbol{\sigma}_\Lambda \cdot \boldsymbol{\sigma}_N)] \left[-F^{(2)}(r) - \frac{2}{r} F^{(1)}(r) \right] \\ & - C_3 \frac{F^{(1)}(r)}{r} \mathbf{L} \cdot \mathbf{S} - C_4 \left[F^{(2)}(r) - \frac{1}{r} F^{(1)}(r) \right] S_{\Lambda N}(\hat{\mathbf{r}}) \\ & - C_5 \frac{F^{(1)}(r)}{r} \mathbf{L} \cdot \mathbf{D}. \end{aligned} \quad (2.2)$$

Here $\mathbf{r}$ is the vector describing the ΛN relative distance, $\boldsymbol{\sigma}_\Lambda$ and $\boldsymbol{\sigma}_N$ are the Pauli matrices representing the Λ and N spin, respectively, $S_{\Lambda N}(\hat{\mathbf{r}}) = 3(\boldsymbol{\sigma}_\Lambda \cdot \hat{\mathbf{r}})(\boldsymbol{\sigma}_N \cdot \hat{\mathbf{r}}) - \boldsymbol{\sigma}_\Lambda \cdot \boldsymbol{\sigma}_N$ is the tensor operator, $\mathbf{L} \cdot \mathbf{S}$ is the spin-orbit operator, with $\mathbf{L}$ the total orbital angular momentum and $\mathbf{S} = \frac{\boldsymbol{\sigma}_\Lambda + \boldsymbol{\sigma}_N}{2}$ the total spin operator, $\mathbf{L} \cdot \mathbf{D}$, being $\mathbf{D} = \frac{\boldsymbol{\sigma}_\Lambda - \boldsymbol{\sigma}_N}{2}$, is the anti-symmetric spin-orbit operator, which is not present in the NN interaction. Furthermore, the functions $F(r)$, $F^{(1)}(r)$ and $F^{(2)}(r)$ are the regulator function and its first and second derivatives, respectively. Their explicit expressions are:

$$F(r) = \frac{1}{R_0^3} \frac{1}{\pi^{3/2}} e^{-\frac{r^2}{R_0^2}}, \quad (2.3)$$

$$F^{(1)}(r) = \frac{1}{\pi^{3/2}} \left(-\frac{2r}{R_0^2} \right) e^{-\frac{r^2}{R_0^2}} \frac{1}{R_0^3}, \quad (2.4)$$

$$F^{(2)}(r) = \frac{1}{\pi^{3/2}} \left(-\frac{2}{R_0^2} + \frac{4r^2}{R_0^4} \right) e^{-\frac{r^2}{R_0^2}} \frac{1}{R_0^3}. \quad (2.5)$$

Here R_0 is the cutoff parameter. We have investigated different values for the cutoff parameter, in the range $R_0 \in [1, 2.5]$ fm. These values were chosen in analogy to Ref. [16]. However, note that the authors of Ref. [16] have used a different Gaussian cutoff, which differentiates between the total isospin channels $T = 0$ and $T = 1$, resulting in a

J^π	α	L	S	$^{2S+1}L_J$
0^+	1	0	0	1S_0
0^-	1	1	1	3P_0
1^+	1	0	1	3S_1
	2	2	1	3D_1
1^-	1	1	0	1P_1
	2	1	1	3P_1

Table 2.6 / Table of α channels for a given total angular momentum and parity J^π , in the ΛN elastic scattering case. Each channel is also labeled in the spectroscopic notation $^{2S+1}L_J$.

cutoff function that is the sum of two Gaussians. In the present case, this distinction does not apply because the isospin of the Λ particle is zero, and thus only one Gaussian appears.

Note also that the model in Eqs. (2.1)–(2.2) still has to be fully determined, as the LECs C_S , C_T , $C_1, \dots, C_5$ have to be fitted to experimental data. Since the available experimental data are Λp elastic scattering cross sections, in the following paragraph we will outline the calculation of the theoretical cross section as a function of the LECs. This procedure was also developed in Ref. [10].

Elastic cross section calculation

The calculation of the elastic Λp cross section requires the solution of the ΛN scattering problem, which means describing the initial and final scattering states. Before proceeding with the description, let us just remark that the elastic scattering process does not result in the creation or destruction of particles. Therefore, the amplitude of the wave function representing the system of the two interacting particles remains unaltered. Only a change in phase of the outgoing wave is possible. This change in phase is referred to as a phase shift δ_L , where L denotes the orbital angular momentum. This quantity is mentioned here for the sake of completeness, as it will be used in the following to describe the scattering process.

In order to describe the initial and final scattering states, it is necessary to employ the total angular momentum J and parity π as quantum numbers. In this study, we included states with total angular momentum $J \leq 1$ and both negative and positive parity. A summary of all allowed states can be found in Table 2.6. It should be noted that, in contrast to the situation involving two nucleons, the wave function does not necessarily have to be anti symmetric. This is due to the fact that we are not dealing with identical particles. Consequently, all possible combinations of orbital angular momentum (L) and spin (S) are permitted. In the notation used in this table, the symbol α represents the channel number, which identifies the specific combination of L, S , given J^π .

To compute the total unpolarized cross section, in Ref. [10] we derived the following expression:

$$\sigma = \frac{4\pi}{k^2} \sum_{J,\alpha,\beta} |T_{\alpha\beta}|^2 \frac{(2J+1)}{(2s_\Lambda+1)(2s_N+1)}, \quad (2.7)$$

where k is the ΛN relative momentum in the center of mass frame, J is the total angular momentum, s_Λ and s_N the Λ and N spin. The indices α and β represent the initial and final channels, respectively, as defined in Table 2.6. The T -matrix encodes the effects of the scattering process, thus, most importantly, it depends on the LECs of the potential model. Note that the T -matrix elements $T_{\alpha\beta}$ in the case of a single channel, reduce to the form $\sin \delta_L e^{i\delta_L}$, where δ_L is the phase shift previously defined.

Experimental data used in the fitting procedure The data available for the fitting procedure consist of cross sections as functions of momentum in the laboratory (LAB) frame. Most are taken from Ref. [21] and cover LAB momenta from 120 MeV to 10^4 MeV. Because we use a contact-interaction potential, only data below a given center-of-mass energy threshold can be included. We set this threshold to 15 MeV at LO and 80 MeV at NLO, with the latter chosen to remain safely below the pion-production threshold. The more restrictive LO cutoff arises from the partial-wave structure of the potential. In the expansion into states of definite orbital angular momentum L , spin S , and total angular momentum J , denoted by $^{2S+1}L_J$, the LO LECs C_S and C_T contribute only to S -waves. Higher partial waves must therefore be excluded from the LO fit. To determine the corresponding E_{CM} threshold, we use a semiclassical estimate

$$L\hbar \simeq b \sqrt{\left[m_\Lambda + \frac{E_{\text{CM}}(E_{\text{CM}} + 2M)}{2m_p} \right]^2 - m_\Lambda^2}, \quad (2.8)$$

where $M = m_\Lambda + m_p$ (m_p is the proton mass and m_Λ is the Λ mass), b is the impact parameter, and E_{CM} is the CM kinetic energy ($c = 1$). Taking $b = 1$ fm and $\hbar c = 197.33$ MeV fm gives $E_{\text{CM}} \simeq 8$ MeV. We nevertheless extend the LO fit to slightly higher energies, up to 15 MeV, because in Ref. [10] the calculated S -wave cross section at this energy was still approximately ten times larger than the higher-partial-wave contributions [10]. Accordingly, at LO we restrict the fit to the positive-parity $J = 0, 1$ channels, corresponding to pure S -waves. At NLO, we include both parities for $J = 0, 1$, thereby allowing P -wave contributions while continuing to exclude higher partial waves.

2.2 New fitting procedure developed in this project

χ^2 function

To determine the values of the LECs, we minimized the following χ^2 function:

$$\chi^2(\mathbf{C}) = \sum_i \frac{[\sigma_i^{th}(\mathbf{C}) - \sigma_i^{exp}]^2}{err(\sigma_i^{exp})^2} + \sum_{j=s,t} \frac{[a_j^{th}(\mathbf{C}) - a_j^{exp}]^2}{err(a_j^{exp})^2}. \quad (2.9)$$

In this expression, σ_i^{th} and σ_i^{exp} represent the theoretical and experimental cross sections for the i -th energy value, respectively, and $err(\sigma_i^{exp})$ denotes the corresponding experimental uncertainty. The LECs are denoted generally as $\mathbf{C}$, where $\mathbf{C} = (C_S, C_T)$ at LO (in this case the number of free parameters N_{par} is 2) and $\mathbf{C} = (C_S, C_T, C_1, \dots, C_5)$ at NLO ($N_{\text{par}} = 7$). The theoretical cross section as a function of the LECs, $\sigma_i^{th}(\mathbf{C})$, is calculated with Eq. 2.7. Note that this χ^2 function includes a second term, which is a sum over the spin-singlet and spin-triplet scattering lengths¹, denoted as a_s and

¹The scattering length is a fundamental parameter in low-energy scattering theory, representing the strength of the scattering process. At very low energies, the total scattering cross section σ is

a_t , respectively (1S_0 and 3S_1 states in spectroscopic notation). Since the available unpolarized cross-section data are weighted sums of the spin-singlet and spin-triplet contributions, they do not by themselves constrain the two S -wave channels independently. Thus, we need the scattering lengths to separately constrain the two S -wave channels. Finally, $a_j^{th}(\mathbf{C})$ is the scattering length obtained with our potential model, while a_j^{exp} and $err(a_j^{exp})$ were obtained from Ref. [23]. No direct measurement of these quantities is available; instead, they are usually inferred from an analysis of scattering cross sections. For this reason, Ref. [23] represents a remarkable result in a new direction, since a_s and a_t were obtained analyzing the femtoscopic Λp correlation functions measured by the ALICE Collaboration. The authors of Ref. [23] obtain a range for a_s and a_t , that is $a_s \in [-3.34, -2.10]$ fm and $a_t \in [-1.56, -1.18]$ fm. Thus in our work we used $a_s^{exp} = -2.72$ fm and $a_t^{exp} = -1.37$ fm, with uncertainties $err(a_s^{exp}) = 0.62$ fm and $err(a_t^{exp}) = 0.19$ fm. This is a new feature of our fitting procedure, since in Ref. [10] we used $a_{s,t}^{exp}$ values derived from theoretical models that were themselves fitted to cross section data, introducing an implicit correlation with the data being fitted here.

Numerical implementation of the fitting procedure

The original fitting code was written entirely in **Fortran** and used the derivative-free **NEWUOA** algorithm from the **NLopt** library [24]. In the present work, we replaced its optimization layer with a **C** routine based on the Toolkit for Advanced Optimization (TAO) in **PETSc** [25]. The scattering observables are still computed by the validated **Fortran** routines, called through a **C**–**Fortran** interface. The physical calculation is therefore unchanged, whereas the minimization, initial conditions, and parallel execution have been redesigned.

The minimization uses **POUNDERS**. Like **NEWUOA**, it is derivative-free and constructs local quadratic models from function evaluations, but it is designed specifically for nonlinear least-squares problems. The function minimized here is the χ^2 defined in Eq. (2.9). In the calculations discussed here: at LO the number of data points N_{data} is $N_{\text{data}} = 13$ and $N_{\text{par}} = 2$, while at NLO $N_{\text{data}} = 24$ and $N_{\text{par}} = 7$.

The workflow of our code is the following: for each value of the cutoff parameter R_0 , an initial set of values for the LECs $\mathbf{C}$ is provided to the **POUNDERS** algorithm. For this set of initial LEC values, the code constructs $V_{\Lambda N}(\mathbf{C})$ and then calculates the cross section for fixed J^π through Eq. (2.7), as well as the scattering lengths. This procedure is repeated for all possible values of J^π , at fixed energy, and then the whole process is repeated for all the E_{CM} values for which there are available data, allowing the computation of the χ^2 value using Eq. (2.9). After the first iteration, the new set of LECs is chosen by **POUNDERS** algorithm and the cycle continues until convergence or the maximum iteration count is reached. Both the convergence threshold and the maximum iteration count are run-time parameters fixed by the user. We performed various convergence tests to determine the optimal values, which is set to 10^{-8} for the convergence threshold and 2000 for the maximum iteration count. Once completed all

related to the scattering length a by $\lim_{k \rightarrow 0} \sigma = 4\pi a^2$, where k is the transferred momentum [22]. The scattering length is a measure of the amplitude of the scattered wave at low energy. The sign of the scattering length also carries physical information. A positive scattering length indicates that the potential supports a bound state near zero energy, while a negative scattering length indicates that the potential does not support such a bound state.

the cycles, the code stores the initial and final LECs, χ^2 , both scattering lengths, and the iteration count.

A local derivative-free minimization can converge to different minima when it is initialized in different regions of a multidimensional parameter space. Rather than relying on a single initial guess, as in Ref. [10], the new implementation defines a grid of N_G starting values, defined as

$$N_G = N_{C_S} N_{C_T} \prod_{i=1}^5 N_{C_i}, \quad (2.10)$$

where N_{C_S} , N_{C_T} , and N_{C_i} are the number of points in the grid for each LEC. To define the grid, an independent minimum, maximum, and spacing can be assigned to each LEC by the user. Every point of the grid is used as the initial condition of an independent POUNDERS minimization.

At LO, both LECs were varied from -15 fm^2 to 15 fm^2 , with a spacing of 1 fm^2 . This gives 31 values for each LEC and therefore

$$N_G^{\text{LO}} = 31^2 = 961 \quad (2.11)$$

independent fits for each value of the regulator R_0 . At NLO, applying the same fine spacing independently to all seven LECs would make the number of starting points computationally prohibitive. We therefore initialize the two LO couplings, C_S and C_T , at the values obtained from the selected best LO fit for the same regulator R_0 . This is also physically well motivated: in an order-by-order EFT expansion, the NLO interaction is expected to provide a correction to the LO result rather than an entirely unrelated solution. Therefore, only the five subleading couplings are gridded at the initialization stage. For each of them we use 11 starting values in the interval $[-15.5, 14.5] \text{ fm}^4$, with a spacing of 3 fm^4 . Hence $N_{C_S} = N_{C_T} = 1$ and $N_{C_i} = 11$, for $i = 1, \dots, 5$, so that the NLO grid contains

$$N_G^{\text{NLO}} = 1^2 \times 11^5 = 161\,051 \quad (2.12)$$

independent starting points for each value of R_0 .

We stress that the LO LECs are fixed only when constructing the set of initial guesses. Once a POUNDERS minimization begins, all seven LECs are treated as free parameters and are allowed to evolve toward the local minimum. The LO-informed initialization therefore reduces the computational cost of the scan without constraining the final NLO solution. Finally, this outer grid scan is parallelized with the Message Passing Interface (MPI), to enhance the code efficiency.

Analysis and selection of the fit results

The grid calculation produces one row for every initial condition and may therefore yield from hundreds to several hundred thousand candidate solutions. We developed a separate Python analysis pipeline to process these large output files.

The metadata stored at the beginning of each file are first read to determine N_{data} , the number of fitted parameters N_{par} , the regulator R_0 , and the numerical settings. The tabulated results are then read in blocks, which avoids keeping the entire NLO file in memory. For every solution we calculate the reduced chi-squared, defined as

$$\chi_\nu^2 = \frac{\chi^2}{\nu}, \quad \nu = N_{\text{data}} - N_{\text{par}}, \quad (2.13)$$

with $\nu = 11$ at LO and $\nu = 17$ at NLO. The upper-tail probability of the chi-squared distribution, corresponding to the conventional goodness-of-fit p -value, is also evaluated. Finally, solutions with positive singlet or triplet scattering length are excluded, since they are not physically meaningful.

At LO, minimizations that started from different grid points (different initial guesses) repeatedly converge to a small number of numerically equivalent minima, *i.e.* usually 3 – 4 combinations of (C_S, C_T) . The solutions can therefore be classified directly according to their final χ_ν^2 . In practice, values of χ_ν^2 are rounded to three decimal places, solutions with the same rounded value are assigned to the same group, and the groups are ordered by their distance from $\chi_\nu^2 = 1$. For each group we record its population and the mean final LECs and scattering lengths. The multiplicity of a group provides a direct measure of how many different initial points reach the same minimum and hence tests the stability of the LO optimization against its starting condition.

At NLO, the situation is more complicated. Seven parameters are constrained by a limited number of observables, and solutions with similar values of χ_ν^2 can correspond to widely separated points in LEC space. Grouping only by the reduced chi-square would therefore mix distinct families of solutions. Therefore, we first require $a_s < 0$ and $a_t < 0$, to select only physically meaningful solutions, and we also require consistency with the selected LO solution at the same value of R_0 , since, as previously discussed, the NLO is expected to be a correction to LO. For this reason, we require that the relative change in the scattering lengths from LO to NLO is less than 30%:

$$\left| \frac{a_s^{\text{NLO}} - a_s^{\text{LO}}}{a_s^{\text{LO}}} \right| < 0.30, \quad \left| \frac{a_t^{\text{NLO}} - a_t^{\text{LO}}}{a_t^{\text{LO}}} \right| < 0.30. \quad (2.14)$$

The surviving NLO solutions are classified in the space of the seven final LECs using the k -means method. Intuitively, this algorithm separates the solutions into k groups by placing nearby points in the same group and iteratively determining the center of each group. In this way, solutions belonging to different regions of the parameter space can be distinguished even when they have similar values of χ_ν^2 . For each cluster, the solution with the smallest χ_ν^2 is selected as the representative set, while the distribution of the other solutions provides an estimate of the range covered by the LECs within that family. The corresponding scattering lengths are also examined, and pairwise projections of the LECs are used to visualize correlations among the parameters.

The resulting ranges should not be interpreted as statistical confidence intervals. The clustering instead identifies distinct, data-compatible families of NLO solutions and summarizes the region of parameter space associated with each family. This is more informative than selecting only the solution with the smallest chi-squared value, which would hide the degeneracy of the NLO fit and its dependence on the initial conditions.

2.3 Results of the fitting procedure

We now present the results obtained with the fitting procedure described in the previous section. We focus on the dependence on the regulator R_0 , on the improvement from LO to NLO, and on the extent to which the fitted interactions can be used outside the energy interval included in the minimization. As discussed in Sec. 2.2, at LO we

R_0 [fm]	LO			NLO		
	χ_ν^2	a_s	a_t	$\min \chi_\nu^2$	range of a_s	range of a_t
		[fm]				[fm]
0.7	0.912	-2.630	-1.115	1.007	[-2.857, -2.834]	[-1.427, -1.407]
1.0	1.105	-3.271	-1.554	1.051	[-2.923, -2.823]	[-1.464, -1.421]
1.5	0.799	-3.044	-1.548	1.021	[-2.819, -2.587]	[-1.440, -1.423]
2.0	3.140	-3.853	-1.929	0.967	[-2.758, -2.714]	[-1.393, -1.371]

Table 2.15 / Summary of the physical LO solutions and of the selected NLO clusters. At LO we report the group with χ_ν^2 closest to one. At NLO the quoted chi-squared is the smallest value among the cluster representatives, while the scattering-length intervals span the cluster means. The NLO ranges describe the multiplicity of data-compatible solutions and are not statistical confidence intervals.

select the physical group whose reduced chi-squared is closest to one. At NLO, several separated families can satisfy the same quality criteria; consequently, we retain the cluster structure rather than assigning a unique solution to each cutoff.

Fit quality and regulator dependence

The main results are summarized in Table 2.15. The LO interaction provides an acceptable description of the harder regulators with $R_0 = (0.7\text{--}1.5)\text{ fm}$, though the extracted scattering lengths exhibit visible cutoff dependence. The quality of the fit becomes worse rapidly at larger R_0 . At $R_0 = 2.0\text{ fm}$, the selected physical solution has $\chi_\nu^2 = 3.140$, and at $R_0 = 2.5\text{ fm}$, which is not reported in the table, it increases to 8.240. Furthermore, only 4.9% and 2.2%, respectively, of the 961 LO minimizations yield negative values for both scattering lengths. These observations suggest that the two LO operators lack the flexibility required for the softer regulators. At NLO the situation changes substantially. For all the values of R_0 shown in Table 2.15, at least one cluster has χ_ν^2 between 0.967 and 1.051. We did not extend the NLO analysis to $R_0 = 2.5\text{ fm}$, since the LO interaction already gives a poor fit at this cutoff, thus making it an unsuitable starting point for an order-by-order improvement. The NLO singlet and triplet scattering lengths lie inside the intervals inferred from the femtoscopic analysis of Ref. [23], namely $a_s \in [-3.34, -2.10]\text{ fm}$ and $a_t \in [-1.56, -1.18]\text{ fm}$. Agreement with these broad intervals is also obtained by the most successful LO solutions; the more significant NLO result is the reduction of the cutoff dependence. Over the cutoffs reported in Table 2.15, the selected LO values span 1.22 fm in a_s and 0.81 fm in a_t , whereas the complete set of NLO cluster means spans only 0.34 fm and 0.09 fm, respectively. Thus, very different sets of NLO LECs lead to nearly the same low-energy scattering lengths.

The coexistence of distinct NLO solutions is illustrated in Table 2.16 for $R_0 = 1.5\text{ fm}$. Both sets of results have comparable fit quality, but several LECs differ in both magnitude and sign. This is not a failure of the minimization: as the last column reports (number of points in each cluster) both regions are repeatedly found. Instead, it is evidence of parameter sloppiness, caused by the limited number and precision of the available unpolarized observables. Additional spin-dependent observables would be required to resolve this degeneracy.

Set	χ_ν^2	C_S [fm ²]	C_T	C_1	C_2	C_3 [fm ⁴]	C_4	C_5	N_{points}
1	1.021	-6.338	4.175	-1.817	2.791	-1.116	0.198	-17.735	533
2	1.112	-15.551	-4.113	-5.901	-2.507	10.802	-0.543	-0.310	667

Table 2.16 / Representative LECs for the two NLO clusters obtained at $R_0 = 1.5$ fm. Each row is the solution with the smallest χ_ν^2 in the corresponding cluster. IN the last column we report the number of points in each cluster, N_{points} .

High-energy predictions

As a final test, we use the fitted interactions to predict the cross section above the energy range used in the minimization. For every value of $R_0 \geq 1.5$ fm analyzed previously, the cross section is calculated using the LECs fitted specifically at that regulator. At each energy, we then take the smallest and largest values among this finite set of predictions as the lower and upper boundaries of the grey envelope (“Theoretical band”) in Fig. 2.17. The band therefore provides a measure of the regulator dependence. It should not be interpreted as a statistical confidence band or as a complete EFT truncation error. The vertical dashed line marks the upper edge of the corresponding fit window.

At LO the different regulators begin to separate soon after the 15 MeV boundary. Although the band follows the observed fall of the cross section up to intermediate momenta, it becomes very broad at high energy. This strong regulator dependence is consistent with the poor LO fit quality found for the softer cutoffs. The NLO band is substantially narrower and remains compatible with most of the data above the 80 MeV fit boundary, including points not used to determine the LECs. The predicted cross section becomes approximately flat, of order 10–20 mb, over a broad momentum interval. The $R_0 = 1.0$ fm curve was not included in this envelope because it develops a large enhancement with the wrong energy dependence above approximately $E_{\text{CM}} = 120$ MeV. A plausible explanation is that this harder regulator suppresses high-momentum components less strongly, the momentum-dependent NLO contact terms can then grow rapidly. This interpretation is not unique, but it is consistent with the fact that the problem appears only beyond the fitted energy range.

The agreement of the remaining curves is encouraging, but it must still be interpreted cautiously. At energies comparable to or larger than the pion-production scale, omitted long-range physics, higher-order contact operators, and additional partial waves can no longer be expected to remain small. The high-energy curves are therefore predictions of the fitted contact model, but not a claim that the NLO EFT is quantitatively controlled throughout the entire range shown.

Overall considerations

In summary, the grid fitting strategy exposes information that would be lost in a single minimization. It shows both the systematic improvement from LO to NLO and the presence of several NLO parameter families. Within the fitted domain, these families lead to stable scattering observables despite their markedly different LECs. Beyond that domain, the reduction of the regulator spread at NLO is promising, while the eventual growth of model dependence sets the natural limit of the contact description.

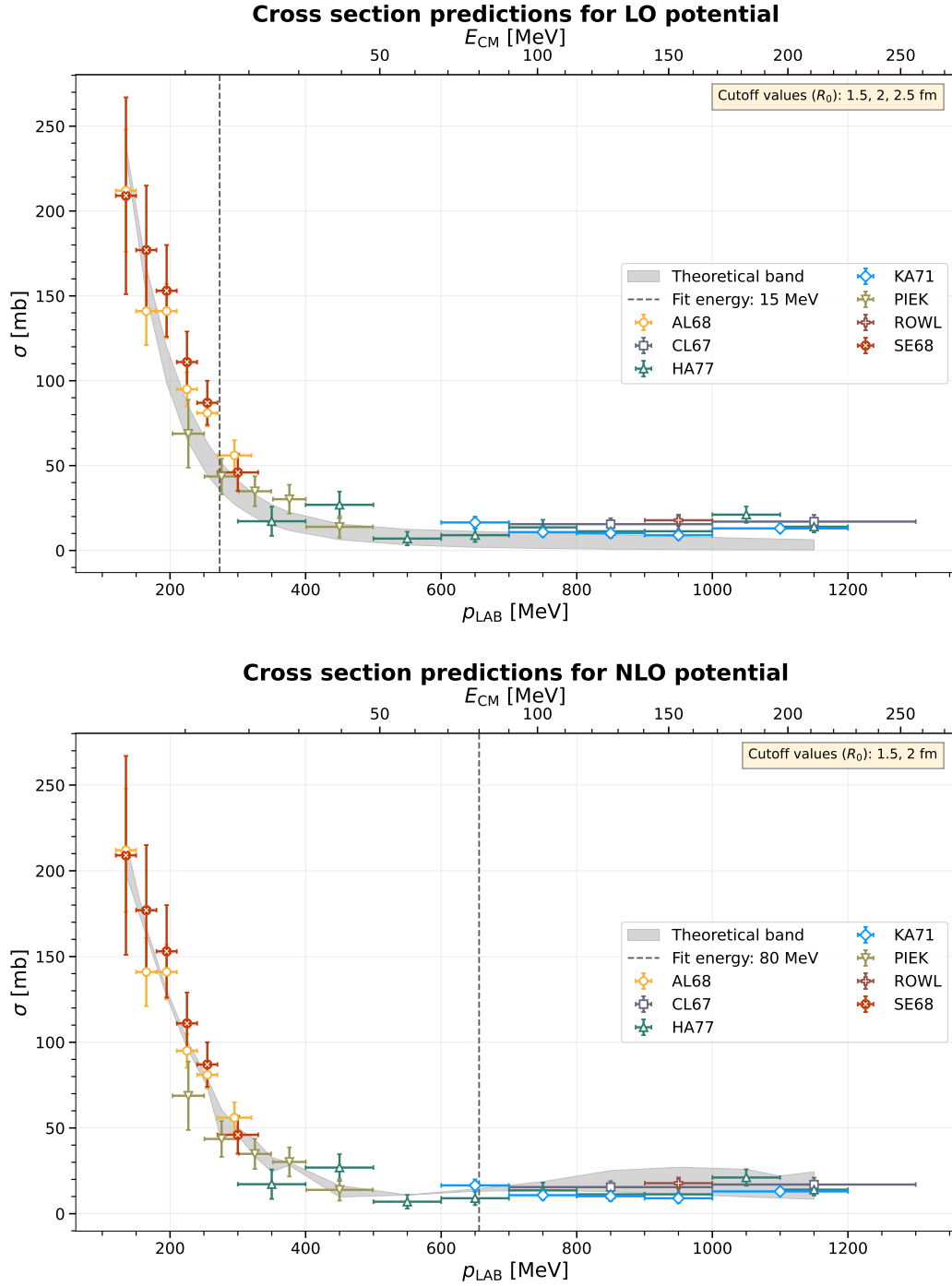


Figure 2.17 / Total Λp cross section predictions as a function of the laboratory momentum p_{LAB} (lower axis) and center-of-mass energy E_{CM} (upper axis). At each energy, the grey bands are bounded by the minimum and maximum cross sections among the discrete R_0 values indicated in each panel; no intermediate regulator values are sampled. The dashed vertical lines mark the maximum energies included in the fits. Data beyond those lines were not used to determine the LECs and provide an out-of-sample comparison. The pathological $R_0 = 1.0$ fm NLO extrapolation is excluded from the envelope, as discussed in the text.

3 Work-in-progress: applications and extensions of the ΛN interaction

The interaction presented in the previous chapters is intended not only to describe two-body scattering, but also to provide the starting point for calculations of hypernuclear observables (for example hypernuclei binding energies) and for a more complete chiral description of the YN force. Both directions are still work in progress: in this chapter we summarize their present status and the main steps that remain, without entering into the technical details of the calculations.

3.1 Application to light hypernuclei

The first application is the calculation of the binding energy of the hypertriton, ${}^3_{\Lambda}\text{H}$ (formed of a proton, a neutron and a Λ hyperon). For this purpose we are using the code developed in Ref. [26], based on the Non-Symmetrized Hyperspherical Harmonics (NSHH) method. In this method, we first separate the center-of-mass motion from the internal dynamics. Then, we expand the internal wave function in hyperspherical harmonics, which generalize ordinary spherical harmonics to the higher-dimensional configuration space of a few-body system. By inserting this expansion into the Rayleigh-Ritz variational principle, the binding energy can be obtained from a eigenvalue-eigenvector problem. In the NSHH formulation, we do not require the nucleon-nucleon wavefunction to be anti-symmetric. The present implementation is specific to three-body systems and it allows the three particles to have non-equal masses, therefore this technique is well suited to a system containing two nucleons and a Λ . As a first step of the present project, the code has been reorganized and optimized to reduce its execution time. This improvement is important because the absence of an explicit anti-symmetrization generally requires a larger basis to reach convergence.

The next validation step is to implement the NN contact interaction of Ref. [16] up to NLO and reproduce the triton ${}^3\text{H}$ binding energy. This interaction has the same local operator structure and regularization scheme (as well as same cutoff parameter values) as the ΛN potential refitted in this work, so that the combination of the two interactions can be performed in a consistent way. We will then combine the NN interaction with the present ΛN interaction and calculate the hypertriton binding energy.

We want to remark here that in Section 2.3 we found that the NLO results led to multiple clusters of solutions with comparable χ^2 . Since we have to choose a single solution, the hypernuclear calculation will be performed with the representative of the cluster whose LECs C_S , C_T (which are the only ones contributing at LO) are closest to the LO solution. As we mentioned earlier, this choice is well motivated by the fact that in an EFT, we expect the NLO contribution to be a subleading correction to LO. However, the ambiguity left by the scattering data will be measured by examining the sensitivity of the binding energy to the other cluster representatives.

3.2 Extension to a local chiral interaction

The second direction of our work aims to extend the contact-only model developed in this thesis to a local χ EFT interaction up to NLO. The important difference is that the new interaction contains not only short-range contact terms, but also the long-range contributions generated by the exchange of pseudoscalar mesons. In the YN sector, the possible exchanged mesons are the π , K , and η , and different baryons can appear in the intermediate states.

The first step is to include the one-pseudoscalar-meson-exchange contribution, which enters at LO. Its momentum-space expression for the YN interaction was derived in Ref. [17]. We Fourier transformed this expression to obtain the corresponding potential in coordinate space. This contribution provides the longest-range part of the interaction.

At NLO, two-pseudoscalar-meson-exchange contributions also appear. Their momentum-space expressions are presented in Ref. [18]. In the YN case, the two exchanged mesons can have different masses. The corresponding expressions must therefore be generalized beyond the equal-mass two-pion case familiar from the NN interaction. For these NLO terms, we use a spectral representation to obtain the coordinate-space potential: it decomposes the complicated two-meson exchange into simple Yukawa-like contributions characterized by an effective exchanged mass μ . For exchanged mesons with masses m_1 and m_2 , μ can take any value from the two-meson threshold $m_1 + m_2$ to infinity. The spectral function determines the weight of each value of μ , and therefore how strongly each corresponding range contributes to the full long-range force. For this transformation, we follow the approach of Ref. [27]. In that work, a fully local chiral NN interaction was derived up to N³LO and applied to calculations of light nuclei. Every result obtained so far in this study has been tested by taking the NN limit and considering pion exchange. In this limit, our expressions reproduce the known results used in Ref. [27].

The YN interaction obtained in this work differs from those of Refs. [17] [18], which are formulated in momentum space and do not lead to a fully local coordinate-space interaction. Our aim is instead to regularize both the contact and meson-exchange contributions locally in coordinate space. A local potential is particularly convenient for computationally demanding few- and many-body calculations. The next step will be to include the long-range one- and two-meson-exchange contributions in the fitting procedure and refit the contact LECs. This will provide the local chiral version of the YN interaction developed in this work.

4 Conclusions and future developments

This work refined the determination of the LECs of the local ΛN contact interaction developed in Ref. [10]. Its derivation and the elastic Λp cross-section calculation were left unchanged, while the goal was a more reliable fit and a more structured pipeline for the analysis of the fitting results.

The new fit combines low-energy Λp cross sections with the spin-singlet and spin-triplet scattering lengths inferred from the femtoscopic analysis of Ref. [23]. Unlike our previous fit, these constraints do not come from theoretical models fitted to the same cross-section data, thus avoiding an implicit correlation. We also replaced the original minimization (with a single initial guess) with a parallel grid of initial conditions. This provides a more systematic exploration of the parameter space and reduces dependence on the initial guess.

At LO, the interaction gives an acceptable description for $R_0 = 0.7\text{--}1.5\text{ fm}$, while the fit becomes progressively worse for the softer regulators. At NLO, at least one solution with $\chi^2_\nu = 0.967\text{--}1.051$ is found for every cutoff considered, $R_0 = 0.7\text{--}2.0\text{ fm}$. The NLO scattering lengths are compatible with the intervals of Ref. [23] and depend much less on the cutoff than the LO results, supporting the expected improvement when additional interaction operators are included. The grid search also revealed distinct NLO families with comparable fit quality. Their LECs can be very different, even though they produce almost identical low-energy observables. Since these families are repeatedly reached from different initial conditions, they reflect an ambiguity left by the limited unpolarized data rather than a failure of the minimization. Additional spin-dependent measurements would be needed to distinguish among them. Above the fitted energy range, the NLO predictions generally show a smaller regulator spread and remain compatible with most of the available data compared with the LO results. This agreement must still be interpreted cautiously, because omitted long-range physics and higher partial waves eventually limit the contact-only description.

Two ongoing projects define the next steps. The first uses the Non-Symmetrized Hyperspherical Harmonics method to apply the refitted interaction to the hypertriton. In the next year, after validation against the triton, the NN and ΛN interactions will be combined to calculate the hypertriton binding energy and test the different fitted families. The second project adds explicit one- and two-pseudoscalar-meson exchange to obtain a local chiral interaction. The derivation of these new terms has already been achieved, and in the next year they will be included in the fitting procedure to obtain the new LECs. This model will then be used for more complex few- and many-body calculations.

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